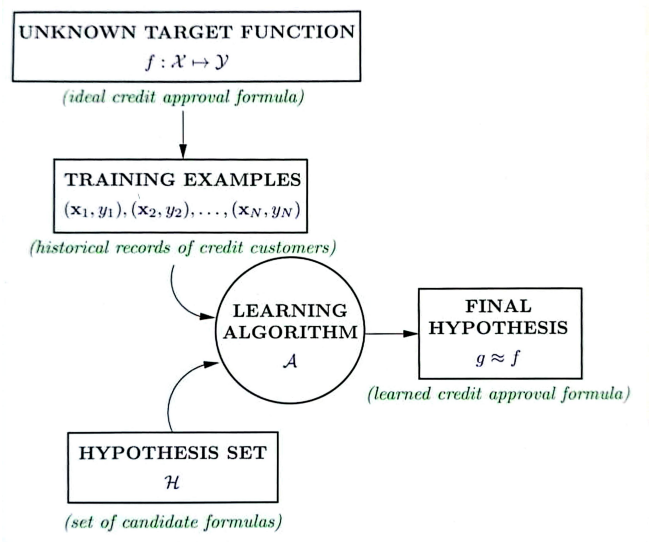
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Semester 2, 2026
Tutorial Week 2

COMP3670 / COMP6670: Introduction to Machine Learning

Question 1 The supervised learning framework

Recall the supervised learning framework from the lecture:



Source: Abu-Mostafa, Y., Magdon-Ismail, M., & Lin, H.-T. (2012). *Learning from Data.* AMLBook. Figure 1.2.

Express each of the following tasks in the framework of learning from data by specifying the input space $\mathcal{X}$, output space $\mathcal{Y}$, target function $f : \mathcal{X} \to \mathcal{Y}$, and the specifics of the dataset that we will learn from.

(a) Medical diagnosis: A patient walks in with a medical history and some symptoms, and you want to identify the problem.

(b) Handwritten digit recognition (for example postal zip code recognition for mail sorting).

(c) Determining if an email is spam or not.

(d) Predicting how an electric load varies with price, temperature, and day of the week.

(e) A problem of interest to you for which there is no analytic solution, but you have data from which to construct an empirical solution.

Solution.

Proposed solution, but other correct solutions exist.

(a) **Medical Diagnosis**

- $\mathcal{X}$: Vector of patient features with age, weight, blood pressure, symptoms (encoded as binary or continuous values), lab test results, etc.

- $\mathcal{Y}$: Set of possible diagnoses (e.g. a disease label, or binary: $\{\text{sick}, \text{healthy}\}$)
- f: The true (unknown) mapping from patient features to the correct diagnosis
- **Data**: Historical patient records where the correct diagnosis has been confirmed by doctors or further testing, pairs $(\mathbf{x}_i, y_i)$

(b) **Handwritten Digit Recognition**

- $\mathcal{X}$: Images of handwritten digits, represented as pixel-intensity vectors, e.g. $\mathcal{X} \subseteq \mathbb{R}^d$ for a d-dimensional flattened image
- $\mathcal{Y}$: $\{0, 1, 2, \ldots, 9\}$
- f: The true mapping from a digit image to the correct digit
- **Data**: A labelled dataset of handwritten digit images with their true digit values

(c) **Spam Detection**

- $\mathcal{X}$: Feature vectors of emails such as word frequencies, presence of keywords, sender metadata
- $\mathcal{Y}$: $\{+1, -1\}$ (spam or not spam)
- f: The true mapping from an email's features to its spam/non-spam label
- **Data**: A collection of emails manually labelled as spam or not spam by users

(d) **Electric Load Prediction**

- $\mathcal{X}$: Tuples of (price, temperature, day of week), so $\mathcal{X} \subseteq \mathbb{R}^2 \times \{\text{Mon}, \ldots, \text{Sun}\}$
- $\mathcal{Y}$: $\mathbb{R}^+$ (load in, say, megawatts)
- f: The true relationship mapping those three variables to the actual electric load
- **Data**: Historical records of (price, temperature, day of week, measured load)

(e) **Open answers**

Question 2 **The Perceptron for Spam Detection**

Suppose that we use a perceptron to detect spam messages. Let's say that each email message is represented by the frequency of occurrence of keywords, and the output is +1 if the message is considered spam.

(a) Write down in mathematical notation the feature vector $\mathbf{x}$ and its domain $\mathcal{X}$. Then write down the formula for the perceptron output $h_{\mathbf{w}}(\mathbf{x})$.

(b) Can you think of some keywords that will end up with a large positive weight in the perceptron?

(c) How about keywords that will get a negative weight?

(d) What parameter in the perceptron directly affects how many borderline messages end up being classified as spam?

Solution.

(a) Each email is represented by the frequency of occurrence of n keywords. The feature vector is

$$\mathbf{x} = (x_1, x_2, \ldots, x_n) \in \mathcal{X},$$

where each $x_i \geq 0$ is the frequency of the i-th keyword, so $\mathcal{X} = \mathbb{R}_+^n$. More precisely, we can take $\mathcal{X} = \mathbb{N}^n$, the n-vectors of natural numbers. The perceptron output is

$$h_{\mathbf{w}}(\mathbf{x}) = \text{sign}\left(\sum_{i=1}^{n} w_i x_i + w_0\right) = \text{sign}(w_1 x_1 + \cdots + w_n x_n + w_0)\,,$$

where $\mathbf{w} = (w_1, \ldots, w_n) \in \mathbb{R}^n$ is the weight vector and $w_0 \in \mathbb{R}$ is the bias. The output is $+1$ if the message is classified as spam and -1 otherwise.

(b) Keywords strongly associated with spam will receive large positive weights, since their high frequency pushes the sum in the perceptron formula above the threshold. Examples include: *prize*, *winner*, *free*, *offer*, *click*, *cash*, *guarantee*, *urgent*.

(c) Keywords that appear frequently in legitimate (non-spam) email will receive negative weights, since their presence should push the output towards -1. Examples include: *meeting*, *lecture*, *assignment*, *regards*, *schedule*, *report*.

(d) The **bias** w_0 (threshold) directly controls how many borderline messages are classified as spam. A more negative bias raises the effective threshold, making the perceptron less likely to classify a message as spam; a more positive bias lowers the threshold, causing more borderline messages to be flagged as spam.

Question 3 **The Perceptron Learning Algorithm**

The **Perceptron Learning Algorithm (PLA)** assumes the dataset is linearly separable and proceeds as follows:

1. **Initialise**: Set $w_0(0) = w_1(0) = \cdots = w_n(0) = 0$.
2. For each iteration $t = 1, 2, \ldots$:
   - Find any misclassified training example $(\mathbf{x}_i, y_i)$ and denote it by $(\mathbf{x}(t), y(t))$ the example selected at step t.
   - **Update** the weights, for $j = 1, \ldots, n$:

$$w_j(t) \leftarrow w_j(t-1) + y(t)\, x_j(t), \qquad w_0(t) \leftarrow w_0(t-1) + y(t). \tag{1}$$

3. **Repeat** step 2 until no misclassified example remains.

The perceptron classifies $\mathbf{x}(t)$ according to $\text{sign}\Big(\sum_{j=1}^{n} w_j(t-1)\, x_j(t) + w_0(t-1)\Big)$, and the weight update rule in (1) has the nice interpretation that it moves in the direction of classifying $\mathbf{x}(t)$ correctly. In this exercise you will show this nice property.

(a) Show that $y(t)\left(\sum_{j=1}^{n} w_j(t-1)\, x_j(t) + w_0(t-1)\right) < 0$. [*Hint*: $\mathbf{x}(t)$ is misclassified by $\mathbf{w}(t-1)$.]

(b) Show that

$$y(t)\left(\sum_{j=1}^{n} w_j(t)\,x_j(t) + w_0(t)\right) > y(t)\left(\sum_{j=1}^{n} w_j(t-1)\,x_j(t) + w_0(t-1)\right).$$

[*Hint*: Use (1).]

(c) As far as classifying $\mathbf{x}(t)$ is concerned, argue that the move from $\mathbf{w}(t-1)$ to $\mathbf{w}(t)$ is a move 'in the right direction'.

For simplification, assume throughout this exercise that $\sum_{j=1}^{n} w_j(t-1)\,x_j(t) + w_0(t-1) \neq 0$ for all t.

Solution.

(a) Since $(\mathbf{x}(t), y(t))$ is misclassified by $\mathbf{w}(t-1)$, the perceptron output disagrees with the true label, i.e. $\operatorname{sign}\left(\sum_{j=1}^{n} w_j(t-1)\,x_j(t) + w_0(t-1)\right) \neq y(t)$. Since $y(t) \in \{-1, +1\}$ and the score is nonzero by assumption, the sum $\sum_{j=1}^{n} w_j(t-1)\,x_j(t) + w_0(t-1)$ and $y(t)$ must have opposite signs, so

$$y(t)\left(\sum_{j=1}^{n} w_j(t-1)\,x_j(t) + w_0(t-1)\right) < 0.$$

(b) Applying the update rule (1), i.e. $w_j(t) = w_j(t-1) + y(t)\,x_j(t)$ for $j = 1, \ldots, n$ and $w_0(t) = w_0(t-1) + y(t)$, we compute:

$$\begin{aligned}\sum_{j=1}^{n} w_j(t)\,x_j(t) + w_0(t) &= \sum_{j=1}^{n}\big(w_j(t-1) + y(t)\,x_j(t)\big)\,x_j(t) + \big(w_0(t-1) + y(t)\big) \\ &= \sum_{j=1}^{n} w_j(t-1)\,x_j(t) + w_0(t-1) + y(t)\sum_{j=1}^{n} x_j(t)^2 + y(t).\end{aligned}$$

Multiplying both sides by $y(t)$ and using $y(t)^2 = 1$:

$$y(t)\left(\sum_{j=1}^{n} w_j(t)\,x_j(t) + w_0(t)\right) = y(t)\left(\sum_{j=1}^{n} w_j(t-1)\,x_j(t) + w_0(t-1)\right) + \sum_{j=1}^{n} x_j(t)^2 + 1.$$

Since $\sum_{j=1}^{n} x_j(t)^2 + 1 > 0$, it follows that

$$y(t)\left(\sum_{j=1}^{n} w_j(t)\,x_j(t) + w_0(t)\right) > y(t)\left(\sum_{j=1}^{n} w_j(t-1)\,x_j(t) + w_0(t-1)\right).$$

(c) Correct classification of $\mathbf{x}(t)$ requires $y(t)\left(\sum_{j=1}^{n} w_j(t)\,x_j(t) + w_0(t)\right) > 0$. From part (a), this quantity is strictly negative before the update (using $\mathbf{w}(t-1)$). From part (b), the update increases it by $\sum_{j=1}^{n} x_j(t)^2 + 1 > 0$, moving it towards a positive value. The move from $\mathbf{w}(t-1)$ to $\mathbf{w}(t)$ therefore always reduces the degree of misclassification of $\mathbf{x}(t)$, which is precisely what we mean by a move 'in the right direction'.